



Airoha IoT SDK for BT Data and Smart Home Power Mode Developer's Guide

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1. Introduction

The system power consumption is a key performance index for user experience. Measurements include the power consumption of data for the MCU, connectivity, and peripheral systems.

- The MCU system includes a processor, memory, and a related clock source.
- The connectivity system includes the baseband, memory, RF, and related clock source.
- The peripheral system usually contains UART, I2C, SPI, GPIO, and more.

This document addresses the MCU system's power mode configuration and power consumption measurement focused on power modes provided by Airoha IoT development platform.

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2. Power Modes and Current Measurement for AB162x

This section introduces the MCU system's power mode and low-power configuration. Power mode for the MCU is described in Section 2.1."Power Mode".

2.1. Power Mode

The power modes for AB162x are summarized below and in Table 1.

- Active
 - The CPU and RAM are in an active state.
 - Instructions can be executed, and peripheral access is active.
- Idle
 - The CPU goes into a clock-gated state to save core power. Internal and external interrupts can wake up the processor.
 - RAM is in an idle state.
- Normal Sleep
 - CPU power is turned off to save more power.
 - RAM enters the lowest power state and the data is preserved.
 - Part of the peripherals are powered off.
- Extended Sleep
 - CPU power is turned off to save more power.
 - Part of the RAM enters the lowest power state and the data is preserved.
 - Part of the RAM enters a power-down state and the data is lost.
 - The size of the retention RAM is configurable.
 - All peripherals, including Always-On-Peripherals, are powered off.
- Deep Sleep
 - Only PMU is retained.
 - RAM goes into a power-down state and the data is lost.

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Table 1. The Power Modes for AB162x

Mode	Description	MCU Clock Source	V _{CORE} Voltage Minimum Value	RAM	Flash	Peripherals
High Speed	The maximum frequency of CM33 is 78 MHz. IBEX is 52 MHz.	LPOSC	0.8V	Active	Active	Active
Full Speed	The maximum frequency of CM33 is 156 MHz. IBEX is 52 MHz.	LPOSC	0.91V			
Idle	The CM33 / IBEX clock is gated to save core power.	Not available	Same as active state	Data kept	Idle	Active
Normal Sleep	The CM33 / IBEX power is turned off to achieve lower current consumption.	Not available	0.8V	Data kept	Idle	Part of the peripherals powered off
Extended Sleep	The CM33 / IBEX power is turned off to achieve lower current consumption.	Not available	Powered off	Part of RAM data kept Part of RAM powered off Size is configurable	Sleep	Powered off
Deep Sleep	Only RTC power is retained.	Not available	Powered off	Powered off (data lost)	Powered off	Powered off

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2.2. Enabling the Airoha Low-Power Mode Based on Zephyr

Airoha low power mode provides a tick suppression option to enable lower power consumption when an idle task is scheduled.

To suspend the ticks, enable the tickless feature in kconfig file and kconfig.sleep_manager, as shown below.

- Enable the airoha tickless feature by “CONFIG_PM” & “CONFIG_PM_POLICY_CUSTOM”

There are two conditions in the tickless porting files (*) that are verified before going into the low-power state. The first condition is that no sleep lock is held by users (**). The second condition is to check if the sleep time is more than 10 ms.

- The system goes into the **Sleep** state if the sleep time is more than 10 ms.
- The system goes into the **Idle** state if the sleep time is less than 10 ms.

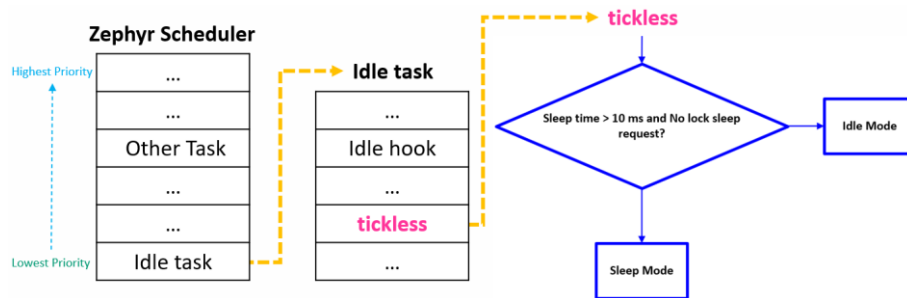


Figure 1. AB162x Airoha Tickless Feature Flow Based on Zephyr

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